

* LSZ-reduction $\Rightarrow$ external states in S-matrix are solutions to free EoMs

off-shell: Correlation f^n

on-shell: S-matrix

- Represent external states as a momentum eigenstate, Eg. ϕ^4 -theory
 $\phi \sim e^{\pm i p x}$ $-$: incoming $+$: outgoing

p : mass-shell condition

- Assume $D=4$ M^4 , external states are massless
 and Assume tree-level, $p^2 = 0$, p^0
 and $p^0 > 0$

- Parametrizing p^2 in 4D $|p^{\alpha\dot{\alpha}} = p^\mu \sigma_\mu^{\alpha\dot{\alpha}}$
 $\Rightarrow$ local isomorphism between

$$\text{Spin}(4, \mathbb{C}) \cong \text{SL}(2, \mathbb{C}) \times \overline{\text{SL}(2, \mathbb{C})}$$

$$\det(p^{\alpha\dot{\alpha}}) = \frac{1}{2} p^2 = 0$$

- Rank 0: trivial

- Rank 1: $p^{\alpha\dot{\alpha}} = \lambda^\alpha \tilde{\lambda}^{\dot{\alpha}}$

$$p^2 = -p_0^2 + p_i^2 = 0$$

- Every null vect. $p^{\alpha\dot{\alpha}} \rightarrow |p\rangle^{\dot{\alpha}} [p]^\alpha$

Replace $\eta_{ab} \rightarrow \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}}$

"Proof": $p^\mu = \frac{1}{2} (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} p_{\alpha\dot{\alpha}}$

$$\begin{cases} \langle k | \gamma^\mu | k \rangle = 2k^\mu \\ \langle k |_{\dot{\alpha}} | \bar{k} \rangle_{\alpha} (\gamma^\mu)^{\alpha\dot{\alpha}} = [k]_{\alpha} \langle k |_{\dot{\alpha}} (\gamma^\mu)^{\alpha\dot{\alpha}} \\ = -k_{\alpha\dot{\alpha}} (\gamma^\mu)^{\alpha\dot{\alpha}} = -(\sigma_{\alpha\dot{\alpha}}^\mu) k_{\alpha} (\gamma^\mu)^{\alpha\dot{\alpha}} \\ = \text{Tr}(\sigma_{\alpha\dot{\alpha}}^\mu) (-k_{\alpha}) = 2k^\mu \quad \checkmark \end{cases}$$

...

$$p^\mu \eta_{\mu\nu} p^\nu = \frac{1}{4} p^{\alpha\dot{\alpha}} \sigma_{\alpha\dot{\alpha}}^\mu \eta_{\mu\nu} \sigma_{\beta\dot{\beta}}^\nu p^{\beta\dot{\beta}}$$

$$\Rightarrow \sigma_{\alpha\dot{\alpha}}^\mu \eta_{\mu\nu} \sigma_{\beta\dot{\beta}}^\nu = -2 \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}}$$

$$\therefore p^{\alpha\dot{\alpha}} \left(-\frac{1}{2} \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} \right) p^{\beta\dot{\beta}} \quad | (\epsilon_{\alpha\beta})_{\text{dynamo}} = -(\epsilon_{\alpha\beta})_{\text{feynman}} \\ \eta_{\mu\nu} \rightarrow -\frac{1}{2} \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}}$$

- Path integral $\Rightarrow$ $\text{Spin}(2) \cong U(1)$

$$\lambda^\alpha = \sqrt{\omega} z^\alpha \quad \bar{\lambda}^{\dot{\alpha}} = \sqrt{\omega} \bar{z}^{\dot{\alpha}} \quad \left| \quad \begin{array}{l} z^\alpha \rightarrow r z^\alpha \\ \bar{z}^{\dot{\alpha}} \rightarrow s \bar{z}^{\dot{\alpha}} \end{array} \right.$$

with ω scaling to compensate / $\omega \rightarrow r^{-1} s^{-1} \omega$

$$\Rightarrow \lambda'^\alpha = \sqrt{\omega'} (z^\alpha)' = r^{-1/2} s^{-1/2} \sqrt{\omega} r z^\alpha = \left(\frac{r}{s}\right)^{1/2} \sqrt{\omega} z^\alpha = \left(\frac{r}{s}\right)^{1/2} \lambda^\alpha$$

$$\tilde{\lambda}'^{\dot{\alpha}} = \left(\frac{s}{r}\right)^{1/2} \tilde{\lambda}^{\dot{\alpha}}$$

$\{[z^\alpha], [\bar{z}^{\dot{\alpha}}]\}$ up to $\mathbb{C}^*$ resulting
 $[z^\alpha]$: coords on $\mathbb{CP}^1 \cong S^2$ ✓

$$z^\alpha = (z^0, z^1) \sim r(z^0, z^1)$$

• ω and a point $(z, \bar{z}) \in S^2 \mid \bar{z} = z^*$

$$\mathcal{U}_0 = \{z^0 \neq 0\}$$

$$\mathcal{U}_1 = \{z^1 \neq 0\}$$

$$\leadsto z^\alpha = z^0 \left(1, \frac{z^1}{z^0}\right)$$

$$z \equiv z^1/z^0, \quad z^\alpha = (1, z)$$



* Let $z = e^{i\phi} \cot(\frac{\theta}{2})$, $(\phi, \theta) \in S^2$

* $(z^\alpha, \bar{z}^{\dot{\alpha}}) \rightarrow (z, \bar{z})$: point on $\mathbb{CP}^1$

$$\hat{\Phi} = \bar{z}^{\dot{\alpha}} p^\alpha \hat{a}_\alpha = \exp\left[\mp i \omega x^{\alpha\dot{\alpha}} z_\alpha \bar{z}_{\dot{\alpha}}\right]$$

$$\begin{aligned} \hookrightarrow p_\mu \chi_\mu &= \frac{1}{2} \langle p | \gamma^\mu | p \rangle \langle x | \gamma_\mu | x \rangle \frac{1}{2} \\ &= \frac{1}{2} \langle p x \rangle [p x] \\ &= -\frac{1}{2} \langle x p \rangle [p | x] \\ &= -\frac{1}{2} x_{\alpha\dot{\alpha}} p^{\alpha\dot{\alpha}} = -\frac{1}{2} x_{\alpha\dot{\alpha}} (\omega z^\alpha \bar{z}^{\dot{\alpha}}) \end{aligned}$$

$$\therefore \hat{\Phi}(x) = a_p^\dagger \exp\left[\mp i p_\mu \chi_\mu\right] = a_p^\dagger \exp\left[\mp \frac{i}{2} \omega x^{\alpha\dot{\alpha}} z_\alpha \bar{z}_{\dot{\alpha}}\right]$$

$$\lambda^\alpha = \sqrt{\omega} z^\alpha$$

$$\bar{\lambda}^{\dot{\alpha}} = \sqrt{\omega} \bar{z}^{\dot{\alpha}}$$

$$\langle \bar{z} | x | z \rangle = \bar{z}_{\dot{\alpha}} x^{\alpha\dot{\alpha}} z_\alpha$$

$$\hat{\Phi}(x) = a_p^\dagger \left[\mp \frac{i}{2} \omega \langle \bar{z} | x | z \rangle \right]$$

• Note : $\langle \bar{z} | x | z \rangle = \langle \bar{z} | x^{\alpha\dot{\alpha}} | z \rangle$

Linearized Gravity : $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu} \quad | \quad \gamma_{\mu\nu} \equiv \kappa h_{\mu\nu}$

Raise and lower with $\eta^{\mu\nu}$

$$g^{\mu\nu} = \eta^{\mu\nu} - \kappa h^{\mu\nu}$$

Work to first order in h , $h_{\mu\nu}(x)$ is a gauge field

$$\Rightarrow \Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\alpha} (\partial_{\mu} g_{\alpha\nu} + \partial_{\nu} g_{\alpha\mu} - \partial_{\alpha} g_{\mu\nu})$$

$$= \frac{1}{2} (\eta^{\rho\alpha} - \kappa h^{\rho\alpha}) (\kappa) (\partial_{\mu} h_{\alpha\nu} + \partial_{\nu} h_{\alpha\mu} - \partial_{\alpha} h_{\mu\nu})$$

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} \eta^{\rho\alpha} \kappa (\partial_{\mu} h_{\alpha\nu} + \partial_{\nu} h_{\alpha\mu} - \partial_{\alpha} h_{\mu\nu})$$

* There are multiple ways to obtain the linear Ricci

$$\text{li) } (R^L)_{\mu\nu} = \delta R_{\mu\nu} = \nabla_{\rho} \delta \Gamma_{\mu\nu}^{\rho} - \nabla_{\mu} \delta \Gamma_{\nu\rho}^{\rho}$$

where $(\Gamma^L)_{\mu\nu}^{\rho} = \delta \Gamma_{\mu\nu}^{\rho}$

$$\text{lii) } \Gamma^2 \text{ terms vanish in } R^L \text{ so } (\Gamma^2 \sim \kappa^2)$$

$$(R^L)_{\mu\nu} = \partial_{\rho} \Gamma_{\mu\nu}^{\rho} - \partial_{\mu} \Gamma_{\nu\rho}^{\rho}$$

* Both identical, but just different ways of viewing them.

$$(R^L)_{\mu\nu} = \frac{1}{2} \kappa \eta^{\rho\alpha} \partial_{\rho} (\partial_{\mu} h_{\alpha\nu} + \partial_{\nu} h_{\alpha\mu} - \partial_{\alpha} h_{\mu\nu}) - \dots$$

$$= \frac{1}{2} \kappa (\partial^{\alpha} \partial_{\mu} h_{\nu\alpha} + \partial^{\alpha} \partial_{\nu} h_{\mu\alpha} - \square h_{\mu\nu})$$

$$- \frac{\kappa}{2} \partial_{\mu} [\eta^{\rho\alpha} (\partial_{\rho} h_{\alpha\nu} + \partial_{\nu} h_{\alpha\rho} - \partial_{\alpha} h_{\rho\nu})]$$

$$= \frac{1}{2} \kappa (\partial^{\alpha} \partial_{\mu} h_{\nu\alpha} + \partial^{\alpha} \partial_{\nu} h_{\mu\alpha} + \square h_{\mu\nu})$$

$$- \frac{\kappa}{2} (\partial_{\mu} \partial^{\alpha} h_{\alpha\nu} + \partial_{\mu} \partial_{\nu} h - \partial_{\mu} \partial^{\rho} h_{\rho\nu})$$

$$= \frac{\kappa}{2} [\partial^{\rho} \partial_{\mu} h_{\rho\nu} - \square h_{\mu\nu} - \partial_{\mu} \partial_{\nu} h + \partial_{\mu} \partial^{\rho} h_{\rho\nu}]$$

$$R_{\mu\nu}^L = \kappa [\partial^{\rho} \partial_{\mu} h_{\rho\nu} - \frac{1}{2} \square h_{\mu\nu} - \frac{1}{2} \partial_{\mu} \partial_{\nu} h]$$

* Linearized Einstein tensor:

$$G_{\mu\nu}^L = R_{\mu\nu}^L - \frac{1}{2} \eta_{\mu\nu} R^L$$

$$G_{\mu\nu}^L = \kappa [\partial^{\rho} \partial_{\mu} h_{\rho\nu} - \frac{1}{2} \square h_{\mu\nu} - \frac{1}{2} \partial_{\mu} \partial_{\nu} h - \frac{1}{2} \eta_{\mu\nu} (\partial^{\rho} \partial^{\sigma} h_{\rho\sigma} - \square h)]$$

Since

$$h^{\mu\nu} R_{\mu\nu}^L = \kappa [\partial^{\rho} \partial^{\mu} h_{\mu\rho} - \square h]$$

* There exists gauge freedom in relativity due to Diffeomorphism: $\psi: M \rightarrow M$ s.t.

$$x^{\mu} \rightarrow x^{\mu} + \kappa \xi^{\mu} \text{ and}$$

$g \sim \psi^* g$ are equivalent, so we really have an equivalence class of metrics. $[g]$

$\Rightarrow h$ and h' are the same physical perturbation iff they differ by an inf. diffeo

$$h' = h + \mathcal{L}_{\xi} h$$

$$\partial_\mu \xi_\nu + (\partial_\mu \xi_\nu)$$

* For $g : (\partial_\xi g)_{\mu\nu} = \kappa (\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu)$

$R : \delta R_{\mu\nu} = (\partial_\xi R)_{\mu\nu} = \kappa (\xi^\rho \nabla_\rho T_{\mu\nu} + \nabla_\mu \xi^\rho R_{\rho\nu} + \nabla_\nu \xi^\rho R_{\rho\mu})$

$\Rightarrow (\partial_\xi h)_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \delta h_{\mu\nu}$
and $\delta R_{\mu\nu}^L = 0$ } To linear order

* Meaning $R_{\mu\nu}^L$ is invariant under gauge transformations
 $h \rightarrow h + 2 \partial_\mu \xi_\nu$

$\therefore$

$$R_{\mu\nu}^L = 0$$

Graviton EOM:

$$\square h_{\mu\nu} - \partial^\rho (\partial_\mu h_{\rho\nu} + \partial_\nu h_{\rho\mu}) + \partial_\mu \partial_\nu h = 0$$

de Donder Gauge: $\partial^\mu h_{\mu\nu} = \frac{1}{2} \partial_\nu h$

• Does not fully fix the gauge since
 $\delta [\partial^\mu h_{\mu\nu} - \frac{1}{2} \partial_\nu h] = \square \xi_\nu$

$\hookrightarrow$ leaves EOM: $\square h_{\mu\nu} = 0$ (Massless scalar)

Lorentz Gauge:

Define the traceless linear term $\tilde{h} = -h$

$$\tilde{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$$

then we can write the linearized Einstein tensor as

$$\begin{aligned} 2 G_{\mu\nu} &= \partial^\rho (\partial_\mu [\tilde{h}_{\rho\nu} + \frac{1}{2} \eta_{\rho\nu} h] + \partial_\nu [\tilde{h}_{\rho\mu} + \frac{1}{2} \eta_{\rho\mu} h]) \\ &\quad - \square (\tilde{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} h) + \partial_\mu \partial_\nu \tilde{h} - \frac{1}{2} \eta_{\mu\nu} [\partial^\rho \partial_\rho (\tilde{h} + \frac{1}{2} \eta h) + \square h] \\ &= \partial^\rho \partial_\mu \tilde{h}_{\rho\nu} - \frac{1}{2} \partial_\nu \partial_\mu \tilde{h} + \partial^\rho \partial_\nu \tilde{h}_{\rho\mu} - \frac{1}{2} \partial_\mu \partial_\nu \tilde{h} \\ &\quad - \square \tilde{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} \square \tilde{h} + \partial_\mu \partial_\nu \tilde{h} - \frac{1}{2} \eta_{\mu\nu} [\partial^\rho \partial_\rho \tilde{h}_{\rho\sigma} - \square \tilde{h}] \\ &= 2 \partial^\rho \partial_\mu \tilde{h}_{\rho\nu} - \partial_\mu \partial_\nu \tilde{h} - \square \tilde{h}_{\mu\nu} + \eta_{\mu\nu} \square \tilde{h} - \frac{1}{2} \eta_{\mu\nu} \partial^\rho \partial_\rho \tilde{h} \\ &\quad + \frac{1}{2} \eta_{\mu\nu} \square \tilde{h} \\ &= 2 \partial^\rho \partial_\mu \tilde{h}_{\rho\nu} - \eta_{\mu\nu} \partial^\rho \partial_\rho \tilde{h}_{\rho\sigma} - \square \tilde{h}_{\mu\nu} \\ &\quad - \partial_\mu \partial_\nu \tilde{h} + \eta_{\mu\nu} \square \tilde{h} + \eta_{\mu\nu} \square \tilde{h} \leftarrow \text{terms} \end{aligned}$$

• Without $\tilde{h} = -h$

$$\begin{aligned}
 2G_{\mu\nu}^L &= \partial^\rho \partial_\mu \tilde{h}_{\nu\rho} + \frac{1}{2} \partial_\nu \partial_\mu h + \partial^\rho \partial_\nu \tilde{h}_{\mu\rho} + \frac{1}{2} \partial_\mu \partial_\nu h \\
 &\quad - \square \tilde{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \square h - \partial_\mu \partial_\nu h - \eta_{\mu\nu} (\partial^\rho \partial_\rho \tilde{h}_{\mu\rho} + \frac{1}{2} \square h - \square h) \\
 &= 2 \partial^\rho \partial_{(\mu} \tilde{h}_{\nu)\rho} - \square \tilde{h}_{\mu\nu} - \eta_{\mu\nu} \partial^\rho \partial_\rho \tilde{h}_{\mu\rho}
 \end{aligned}$$

$$2G_{\mu\nu}^L = 2 \partial^\rho \partial_{(\mu} \tilde{h}_{\nu)\rho} - \square \tilde{h}_{\mu\nu} - \eta_{\mu\nu} \partial^\rho \partial_\rho \tilde{h}_{\mu\rho} = 0$$

* Since $\tilde{h}$ and h are equivalent, then $\exists \xi^\mu$ s.t.

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$$

$$h \rightarrow h'_{\mu\nu} = h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

$$h' = h + 2 \partial^\mu \xi_\mu$$

$$\Rightarrow \tilde{h}'_{\mu\nu} = h'_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h' = h'_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} (h + 2 \partial^\rho \xi_\rho)$$

$$= h'_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h - \eta_{\mu\nu} \partial^\rho \xi_\rho$$

$$= h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \frac{1}{2} \eta_{\mu\nu} h - \eta_{\mu\nu} \partial^\rho \xi_\rho$$

$$\tilde{h}'_{\mu\nu} = \tilde{h}_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \eta_{\mu\nu} \partial^\rho \xi_\rho$$

$$\partial^\nu \tilde{h}'_{\mu\nu} = \partial^\nu \tilde{h}_{\mu\nu} + \partial_\mu (\partial^\nu \xi_\nu) + \square \xi_\mu - \partial_\mu (\partial^\rho \xi_\rho)$$

$$= \partial^\nu \tilde{h}_{\mu\nu} + \square \xi_\mu$$

* Now choose? $\partial^\nu \tilde{h}'_{\mu\nu} = 0$ so that

$$\left\{ \begin{aligned} &\forall x \in M: \square \chi = -\partial_\mu A^\mu = 0 \end{aligned} \right.$$

$$-\partial^b \tilde{h}_{ab} = \square \xi_b$$

$$\bullet \tilde{h}' \rightarrow \tilde{h} \text{ as } \tilde{h} \rightarrow h'$$

$$\Rightarrow \boxed{\partial^b \tilde{h}_{ab} = 0}$$

Means the linearized Einstein eqs become

$$\boxed{\square \tilde{h}_{\mu\nu} = 0}$$

* Choose the non-covariant gauge condition:

$$\boxed{\partial^i h_{im} = 0}$$

1) This is diffeomorphism invariant:

$$\delta(\partial^i h_{im}) = \partial^i (\partial_i \xi_m + \partial_m \xi_i)$$

$$= \nabla^2 \xi_m + \partial_m (\partial^i \xi_i)$$

* We require this to be invariant:

hence

$$\delta(\partial^i h_{im}) = 0$$

$$\nabla^2 \xi_m + \partial_m (\partial^i \xi_i) = 0$$

$$\underline{j}: \nabla^2 \xi_j + \partial_j (\partial^i \xi_i) = 0$$

$$\nabla^2 (\partial^i \xi_i) + \nabla^2 (\partial^i \xi_i) = 0$$

$$\Rightarrow \partial^j \xi_j = 0$$

$$\Rightarrow \nabla^2 \xi_m = 0 \Rightarrow \xi_m(x) \equiv 0$$

$$R_{\mu\nu}^L = 0 = \square h_{\mu\nu} - 2 \partial^\rho (\partial_\mu h_{\nu\rho}) + \partial_\mu \partial_\nu h$$

$$\square h_{\mu\nu} - 2 \partial^\rho \partial_\mu h_{\nu\rho} + \partial_\mu \partial_\nu h = 0$$

$$\square h_{\mu\nu} - 2 \partial^\rho \partial_\mu h_{\nu\rho} + \partial_\mu \partial_\nu (h_{ii} - h_{00}) = 0$$

$$\underline{00}: \square h_{00} + 2 \partial_0^2 h_{00} + \partial_0^2 (h_{ii} - h_{00}) = 0$$

$$\nabla^2 h_{00} + \partial_0^2 h_{ii} = 0$$

$$\underline{ii}: \square h_{ii} + 2 \partial_0 \partial_i h_{i0} + \partial_i^2 (h_{ii} - h_{00})$$

$$- \partial_i^2 h_{00} + 2 \nabla^2 h_{ii} + 2 \partial_0 \partial_i h_{i0} - \nabla^2 h_{00} = 0$$

$$\underline{i0}: \square h_{i0} + \partial_0 (\partial_i h_{00} + \partial_0 h_{i0}) + \partial_i \partial_0 (h_{ii} - h_{00})$$

$$= -\partial_0^2 h_{i0} + \nabla^2 h_{i0} + \partial_0 \partial_i h_{00} + \partial_0^2 h_{i0} +$$

$$\partial_i \partial_0 h_{ii} - \partial_i \partial_0 h_{00} = 0$$

$$\Rightarrow \nabla^2 h_{i0} + \partial_i \partial_0 h_{ii} = 0$$

$\hookrightarrow \partial^i h_{i0} = 0$ thus from variables

$$\underline{ii}: -\partial_0^2 h_{ii} + 2 \nabla^2 h_{ii} - \nabla^2 h_{00} = 0$$

* Adding $00 + ii$:

$$\sum_i \nabla^2 h_{ii} = 0 \Rightarrow \sum_i h_{ii} = 0$$

$$\Rightarrow \nabla^2 h_{00} = 0 \Rightarrow h_{00} = 0$$

$$\underline{i0}: \square h_{i0} + \partial_0^2 h_{i0} = 0$$

$$\Rightarrow \nabla^2 h_{i0} = 0 \Rightarrow h_{i0} = 0$$

* Thus, only h_{ij} survives

$$\underline{ij}: \square h_{ij} = 0$$

So h_{ij} has $(D-1)(D-2)/2$ components
subject to

(i) Trace: $h_{ii} = 0$ (1)

(ii) Gauge: $\partial^i h_{ij} = 0$, (D-1)

$$\rightarrow h_{ij} = (D-1)^2 - (D-1) = (D-1)$$

2

$$= ((D-1)^2 + (D-1))/2 = (D-1)[(D-1)+1]/2$$

$$|h_{ij}| = D(D-1)/2$$

$$D(D-1)/2 - (D-1) - 1 = \boxed{\frac{D(D-3)}{2}} \quad \text{D.O.F. of } \text{SO}(D-2)$$

* Gauge invariant content of a weak gravitational field lies in the Weyl components of the linearized Riemann tensor:

$$R_{\mu\nu\rho\sigma}^L = C_{\mu\nu\rho\sigma}^L$$

$$\Rightarrow \boxed{\square R_{\mu\nu\rho\sigma}^L = 0}$$

Gravitational Green's function:

$$-\square h_{\mu\nu} + \partial^\rho (\partial_\mu h_{\rho\nu} + \partial_\nu h_{\rho\mu}) - \partial_\mu \partial_\nu h = 2\kappa (T_{\mu\nu} - \frac{1}{D-2} \eta_{\mu\nu} T)$$

$$\boxed{h_{\mu\nu}(x) = \kappa \int d^D y G_{\mu\nu;\mu'\nu'}(x) T^{\mu'\nu'}(y)}$$

$$x \equiv (x-y)^2$$

* x^{μ} and $y^{\mu'}$
 $\hookrightarrow$ observation point $\hookrightarrow$ source point

* Under a change of coordinates $x \rightarrow \tilde{x}$, consider only global Lorentz. G is a bi-tensor

$$\kappa \int d^D y (\square G T + \partial^\rho \partial_\mu G T + \partial^\rho \partial_\nu G T - \partial_\mu \partial_\nu G T) = 2\kappa (T_{\mu\nu} - \frac{1}{D-2} \eta_{\mu\nu} T)$$

RHS:

$$2\kappa (T_{\mu\nu}(x) - \frac{1}{D-2} \eta_{\mu\nu} T(x)) = 2\kappa \int d^D y \delta(x-y) (\quad)$$

$$T_{\mu\nu} = \frac{1}{2} \eta_{\mu\mu'} T^{\mu'\nu'} \eta_{\nu\nu'} + \frac{1}{2} (\leftrightarrow)$$

$$\kappa (\eta_{\mu\mu'} \eta_{\nu\nu'} + \eta_{\mu\nu'} \eta_{\nu\mu'}) T^{\mu'\nu'}$$

* For the trace: $\eta_{\mu'\nu'} T^{\mu'\nu'} = \eta_{\mu\nu} T^{\mu\nu}$

$$2\kappa \int d^D x \delta^D(x-y) \left[\eta_{\mu\mu'} \eta_{\nu\nu'} + \eta_{\mu\nu'} \eta_{\nu\mu'} - \frac{2}{D-2} \eta_{\mu\nu} \eta_{\mu'\nu'} \right] T^{\mu\nu}$$

+ Boundary terms

$$\hookrightarrow \partial_{\mu'} \Delta_{\mu\nu;\nu'} + \partial_{\nu'} \Delta_{\mu\nu;\mu'}$$

* Example Maxwell:

$$(\delta^{\mu\rho} \square - \partial^\mu \partial^\rho) G_{\mu\nu}(x,y) = -\delta^\mu_\nu \delta^D(x-y)$$

but $(\delta^{\mu\rho} \square - \partial^\mu \partial^\rho) = \square^{\mu\rho}$ is non-invertible
since $\forall f \quad \partial_\rho f$ is a zero mode

$$(\delta^{\mu\rho} \square - \partial^\mu \partial^\rho) \partial_\rho f = \partial^\mu \square f - \partial^\mu \square f = 0$$

* For any^{elect.} source J : $\square F_{\mu\rho} = -(\partial_\nu J_\rho - \partial_\rho J_\nu)$
 $\hookrightarrow$ Impose conditions on Ω

* Here, LHS is also non-invertible. Δ are pure gauge

* Symmetries + Gauge:

$$G_{\mu\nu\mu'\nu'}(z) = (\eta_{\mu\mu'} \eta_{\nu\nu'} + \eta_{\mu\nu'} \eta_{\nu\mu'}) G(z) + \eta_{\mu\nu} \eta_{\mu'\nu'} H(z) + (3 \text{ other terms})$$

$$\hookrightarrow C(z) (\eta_{\mu\mu'} z_\nu z_{\nu'} + \eta_{\mu\nu'} z_\nu z_{\mu'} + \eta_{\nu\mu'} z_\mu z_{\nu'} + \eta_{\nu\nu'} z_\mu z_{\mu'})$$

$$+ D(z) (\eta_{\mu\nu} z_{\mu'} z_{\nu'} + \eta_{\mu'\nu'} z_\mu z_\nu)$$

$$+ E(z) z_\mu z_\nu z_{\mu'} z_{\nu'}$$

* Couples only to conserved T : $\partial^\mu T_{\mu\nu} = 0$

$\hookrightarrow$ Plugging G in:

$$\square G(z) = -\delta^D(x-y)$$

$$H(z) = -\frac{2}{D-2} G(z)$$

$\Rightarrow$

$$G_{\mu\nu\mu'\nu'}(k) = (\eta_{\mu\mu'} \eta_{\nu\nu'} + \eta_{\mu\nu'} \eta_{\nu\mu'} - \frac{2}{D-2} \eta_{\mu\mu'} \eta_{\nu\nu'}) G(k) + (\text{Gauge})$$

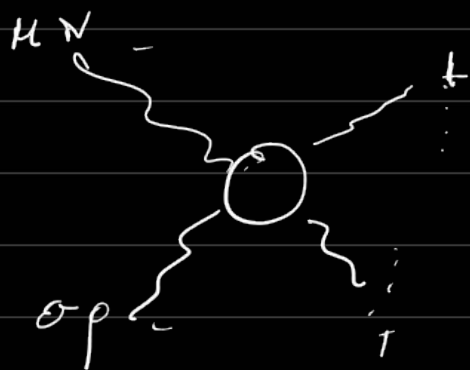
$$-k^2 G(k) = -1$$

$$\Rightarrow G(k) = \frac{1}{k^2}$$

$$G_{\mu\nu\mu'\nu'}(k) = (\eta_{\mu\mu'} \eta_{\nu\nu'} + \eta_{\mu\nu'} \eta_{\nu\mu'} - \frac{2}{D-2} \eta_{\mu\mu'} \eta_{\nu\nu'}) \frac{1}{2\kappa k^2}$$

* For each rank-2 tensor, there are $\frac{D(D+1)}{2} - 1$ independent components

* each graviton leg is labeled by two wavevectors.



* Dot in polarization vectors for each external line
4D

$$\epsilon_{-}^{\mu\nu}(p_i) = \epsilon_{-}^{\mu}(p_i) \epsilon_{-}^{\nu}(p_i)$$

$$\epsilon_{+}^{\mu\nu}(p_i) = \epsilon_{+}^{\mu}(p_i) \epsilon_{+}^{\nu}(p_i)$$

$$V_3(p_1, p_2, p_3) = p_1^{\mu_1} p_2^{\nu_2} \eta^{\mu_1 \nu_2} \eta^{\mu_2 \nu_1} + \dots$$

$$M_4^{\text{tree}}(1^- 2^- 3^+ 4^+) = \frac{\langle 12 \rangle^7 [12]}{s t u}$$

$$= \langle 13 \rangle \langle 14 \rangle \langle 23 \rangle \langle 24 \rangle \langle 34 \rangle^2$$

$$= \frac{\langle 12 \rangle^4 [34]^4}{s t u}$$

• KLT-relations etc.

$$3\text{pt. } M_3(1^- 2^- 3^+) = A_3[1^- 2^- 3^+]^2$$

$$M_3$$

